

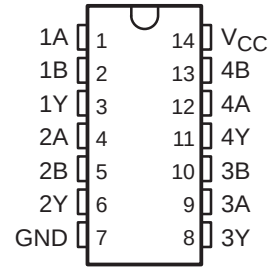
SN54HC00, SN74HC00 QUADRUPLE 2-INPUT POSITIVE-NAND GATES

SCLS181B – DECEMBER 1982 – REVISED MAY 1997

- Package Options Include Plastic Small-Outline (D), Thin Shrink Small-Outline (PW), and Ceramic Flat (W) Packages, Ceramic Chip Carriers (FK), and Standard Plastic (N) and Ceramic (J) 300-mil DIPs

数据手册修订日期，在使用前注意核对一下日期，以保证获得最新的信息

SN54HC00 ... J OR W PACKAGE
SN74HC00 ... D, N, OR PW PACKAGE
(TOP VIEW)



同为74HC00但是有多种封装形式，可通过末尾的字母来区分

description

These devices contain four independent 2-input NAND gates. They perform the Boolean function $Y = A \cdot B$ or $Y = \overline{A + B}$ in positive logic.

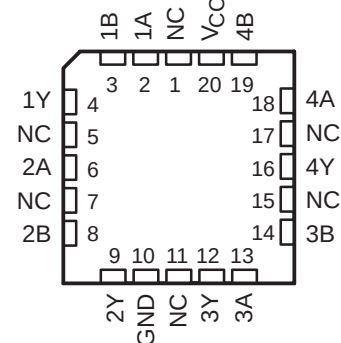
The SN54HC00 is characterized for operation over the full military temperature range of -55°C to 125°C . The SN74HC00 is characterized for operation from -40°C to 85°C .

FUNCTION TABLE
(each gate)

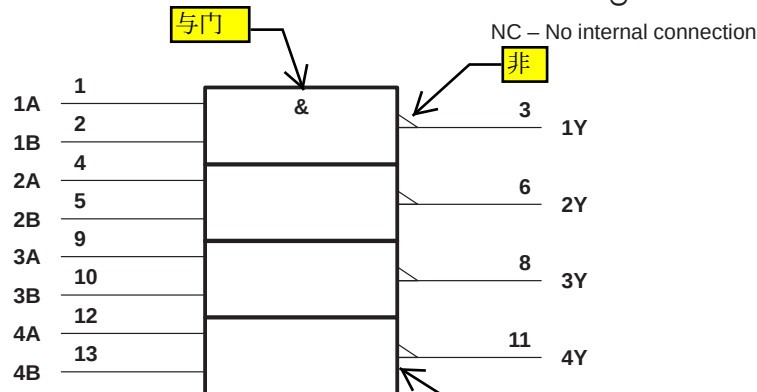
INPUTS		OUTPUT
A	B	Y
H	H	L
L	X	H
X	L	H

逻辑功能表

SN54HC00 ... FK PACKAGE
(TOP VIEW)



logic symbol†

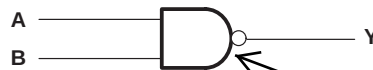


† This symbol is in accordance with ANSI/IEEE Std 91-1984 and IEC Publication 617-12. Pin numbers shown are for the D, J, N, PW, and W packages.

图形符号说明

图形符号，含逻辑和管脚信号，该符号由ANSI/IEEE Std 91-1984 and IEC Publication 617-12标准规定

logic diagram (positive logic)



逻辑符号



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PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

TEXAS
INSTRUMENTS

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极限工作参数列表，给出什么条件将毁坏芯片，注意不是最大工作限制

4HC00 INPUT POSITIVE-NAND GATES

看注意事项1
获取详细信

输入钳位电流，指在正常工作电压范围之外可安全流过器件输入终端的最大电流

输出钳位电流，指在正常工作电压范围之外可安全流过器件输出终端的最大电流

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absolute maximum ratings over operating free-air temperature range†

最大电源电压范围

Supply voltage range, V_{CC}	–0.5 V to 7 V
Input clamp current, I_{IK} ($V_I < 0$ or $V_I > V_{CC}$) (see Note 1)	±20 mA
Output clamp current, I_{OK} ($V_O < 0$ or $V_O > V_{CC}$) (see Note 1)	±20 mA
Continuous output current, I_O ($V_O = 0$ to V_{CC})	±25 mA
Continuous current through V_{CC} or GND	±50 mA
Package thermal impedance, θ_{JA} (see Note 2): D package	127°C/W
N package	78°C/W
PW package	170°C/W
Storage temperature range, T_{stg}	–65°C to 150°C

最大连续输出电流

最大连续电源电流

器件封装热阻抗，指器件消耗1W功率引起的温度升高值，不同封装形式是不一样的

† Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

NOTES: 1. The input and output voltage ratings may be exceeded if the input and output current ratings are observed.
2. The package thermal impedance is calculated in accordance with JESD 51, except for through-hole packages, which use a trace length of zero.

推荐工作条件

recommended operating conditions

不同电源电压条件下，参数值是不一样的

参数值有最小、典型和最大之分

			SN54HC00			SN74HC00			UNIT		
			MIN	NOM	MAX	MIN	NOM	MAX			
V _{CC}	Supply voltage		2	5	6	2	5	6	V		
V _{IH}	High-level input voltage 输入高电平电压	V _{CC} = 2 V	1.5			1.5			V		
		V _{CC} = 4.5 V	3.15			3.15					
		V _{CC} = 6 V	4.2			4.2					
V _{IL}	Low-level input voltage 输入低电平电压	V _{CC} = 2 V	0			0			V		
		V _{CC} = 4.5 V	0			1.35					
		V _{CC} = 6 V	0			1.8					
V _I	Input voltage	输入电压范围		0	V _{CC}		0	V _{CC}		V	
V _O	Output voltage			0	V _{CC}		0	V _{CC}		V	
t _t	Input transition (rise and fall) time 输入过渡时间（上升或下降时	V _{CC} = 2 V	0			1000			ns		
		V _{CC} = 4.5 V	0			500					
		V _{CC} = 6 V	0			400					
T _A	Operating free-air temperature	输入电压范围		−55	125		−40	85		°C	

SN54HC00, SN74HC00 QUADRUPLE 2-INPUT POSITIVE-NAND GATES

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electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		V _{CC}	T _A = 25°C			SN54HC00		SN74HC00		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
V _{OH} 输出高电平电压	V _I = V _{IH} or V _{IL}	I _{OH} = -20 µA	2 V	1.9	1.998		1.9		1.9		V
			4.5 V	4.4	4.499		4.4		4.4		
			6 V	5.9	5.999		5.9		5.9		
		I _{OH} = -4 mA	4.5 V	3.98	4.3		3.7		3.84		
		I _{OH} = -5.2 mA	6 V	5.48	5.8		5.2		5.34		
V _{OL} 输出低电平电压	V _I = V _{IH} or V _{IL}	I _{OL} = 20 µA	2 V		0.002	0.1		0.1		0.1	V
			4.5 V		0.001	0.1		0.1		0.1	
			6 V		0.001	0.1		0.1		0.1	
		I _{OL} = 4 mA	4.5 V		0.17	0.26		0.4		0.33	
		I _{OL} = 5.2 mA	6 V		0.15	0.26		0.4		0.33	
I _I	V _I = V _{CC} or 0	单个输入端输入电流	6 V		±0.1	±100		±1000		±1000	nA
I _{CC}	V _I = V _{CC} or 0, I _O = 0		6 V			2		40		20	µA
C _i	输入等效容抗		2 V to 6 V		3	10		10		10	pF

switching characteristics over recommended operating free-air temperature range, C_L = 50 pF (unless otherwise noted) (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V _{CC}	T _A = 25°C			SN54HC00		SN74HC00		UNIT
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
t _{pd} 转换时间, 定义 请参看后文	A or B	Y	2 V		45	90		135		115	ns
			4.5 V		9	18		27		23	
			6 V		8	15		23		20	
t _t 过渡时间 (上升 或下降时间)		Y	2 V		38	75		110		95	ns
			4.5 V		8	15		22		19	
			6 V		6	13		19		16	

operating characteristics, T_A = 25°C

PARAMETER				TEST CONDITIONS	TYP	UNIT
C _{pd}	Power dissipation capacitance per gate	每个门的动态功耗等效容抗		No load	20	pF

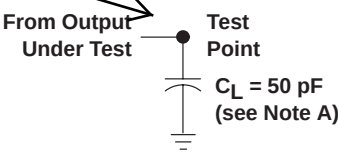
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QUADRUPLE 2-INPUT POSITIVE-NAND GATES

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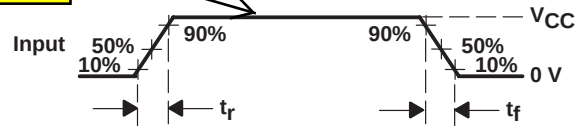
动态开关参数定义

动态开关参数的测试电路

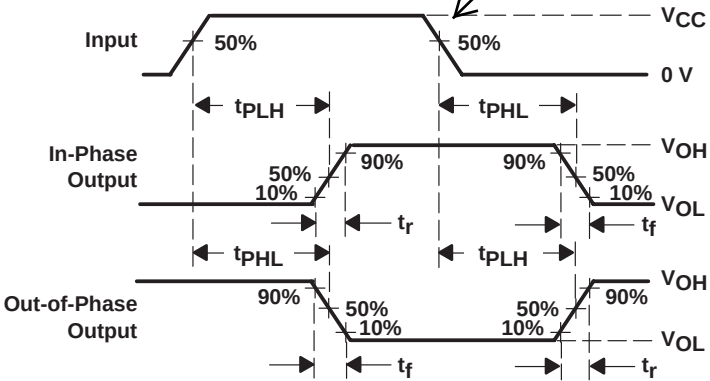
PARAMETER MEASUREMENT INFORMATION



LOAD CIRCUIT



VOLTAGE WAVEFORM
INPUT RISE AND FALL TIMES



VOLTAGE WAVEFORMS
PROPAGATION DELAY AND OUTPUT TRANSITION TIMES

- NOTES:
- A. C_L includes probe and test-fixtue capacitance.
 - B. Phase relationships between waveforms were chosen arbitrarily. All input pulses are supplied by generators having the following characteristics: $PRR \leq 1 \text{ MHz}$, $Z_O = 50 \Omega$, $t_r = 6 \text{ ns}$, $t_f = 6 \text{ ns}$.
 - C. The outputs are measured one at a time with one input transition per measurement.
 - D. t_{PLH} and t_{PHL} are the same as t_{pd} .

Figure 1. Load Circuit and Voltage Waveforms

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